

Types of network: zoom out

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 1.3 Computer networks, connections and protocols — Paper 1, Computer systems
Lesson	Types of network: zoom out (lesson 1 of the unit)
Lesson ID	nw-101
Experience ID	nw-101
Plan ID	PLAN-NW-L01
Content version	acd77acd5d9b
Website	https://revise360.co.uk/experience.html?id=nw-101

Big question, outcomes and success criteria

Not in the lesson record: a key question for this lesson. Add it to the lesson source and rebuild this plan rather than writing it on the printout.

Prerequisites

None within this unit: it is the first lesson. Treat the starter as a baseline check rather than retrieval of something you have taught.

Assessable outcomes

1. **nw-101-o1** — know what is meant by a standalone computer
2. **nw-101-o2** — know the different types of network: LAN and WAN
3. **nw-101-o3** — understand the advantages of networking
4. **nw-101-o4** — understand the implications (disadvantages) of networking

Vocabulary, resources and preparation

Not in the lesson record: key vocabulary. Add it to the lesson source and rebuild this plan rather than writing it on the printout.

Resources

SLIDES	presentations/NW_L01_TypesOfNetworkZoomOut_Lesson.pptx
WORKSHEET	worksheets/NW_L01_TypesOfNetworkZoomOut_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=nw-101

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on <code>NW_L01_TypesOfNetworkZoomOut_Worksheet.docx</code> is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find Standalone computer.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 15 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exit questions a) to d)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. Standalone computer

- A computer that is not connected to any other computer or network.
- Its files, software and hardware can only be used on that one machine.

Challenge on this wall: Name one device in your home that is standalone and one that is on a network.

2. Sharing a file

- Without a network, files must be copied onto removable media, such as a USB stick.
- The stick is then carried to the other computer and copied again.

Challenge on this wall: You and a friend both edit the same essay by passing a USB stick back and forth. What could go wrong?

3. Printing

- Each standalone computer needs its own printer, plugged in directly.
- A room of 30 standalone computers could need 30 printers.

Challenge on this wall: Why would this be a problem for a school's budget?

4. Updates and backups

- Software must be installed and updated on every computer separately.
- Each computer has to be backed up on its own, which is easy to forget.

Challenge on this wall: This computer's hard drive fails tonight and it has never been backed up. What is lost?

6. Local area network (LAN)

- Covers a small geographical area, such as one building or one school site.
- The organisation owns and manages all of the hardware.
- Devices connect with cables or Wi-Fi.

Challenge on this wall: Is the Wi-Fi network in your home a LAN? Use the definition to explain.

7. Sharing resources

- Files saved on a file server can be opened from any computer.
- One printer can be shared by a whole room.
- You can log on at any computer and see your own files.

Challenge on this wall: Name three things in your school that are shared over the LAN.

8. Advantages

- Share files and hardware, such as printers.
- Install software and updates on every computer from one place.
- Back up everyone's data centrally.
- Communicate using email and messaging.

Challenge on this wall: Which advantage do you think saves the school the most money? Justify your choice.

9. Disadvantages

- Viruses and malware can spread from one computer to many.
- Servers, switches and cabling are expensive.
- A network manager is needed to run it.
- If the file server fails, nobody can reach their files.

Challenge on this wall: How could a school reduce the risk of a virus spreading across its network?

11. Wide area network (WAN)

- Covers a large geographical area: across towns, countries or the whole world.
- Connects LANs together.
- The internet is the biggest WAN.

Challenge on this wall: A school trust links the LANs of its schools in different towns. Is that a LAN or a WAN? Why?

12. Who owns the connections?

- WAN links are usually owned by telecoms companies. Organisations rent (lease) them.
- Links include fibre optic cables (some under the sea), phone lines and satellites.

Retrieval and the check for understanding

Not in the lesson record: a starter to retrieve prior knowledge. Add it to the lesson source and rebuild this plan rather than writing it on the printout.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (nw-101-s8-c1), so the wording a pupil sees matches the wording you say.

Which of these is an advantage of a network?

Correct: Software can be installed and updated on every computer from one place

What a pupil might choose instead

- Viruses can spread from one computer to others
- A network manager is needed to run it
- Servers and switches are expensive

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: The other three options are all disadvantages of networks.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	26	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	1	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	Standalone computer	2	Name one device in your home that is standalone and one that is on a network.
2	Sharing a file	2	You and a friend both edit the same essay by passing a USB stick back and forth. What could go wrong?
3	Printing	2	Why would this be a problem for a school's budget?
4	Updates and backups	2	This computer's hard drive fails tonight and it has never been backed up. What is lost?
5	Predict	1	the answer to the station's challenge
6	Local area network (LAN)	2	Is the Wi-Fi network in your home a LAN? Use the definition to explain.
7	Sharing resources	2	Name three things in your school that are shared over the LAN.
8	Advantages	2	Which advantage do you think saves the school the most money? Justify your choice.
9	Disadvantages	2	How could a school reduce the risk of a virus spreading across its network?
10	The LAN in this room	2	the answer to the station's challenge
11	Wide area network (WAN)	2	A school trust links the LANs of its schools in different towns. Is that a LAN or a WAN? Why?
12	Who owns the connections?	3	Why doesn't a school lay its own cable to another school 30 miles away?
13	LAN vs WAN	1	the answer to the station's challenge
14	Quick check	1	the answer to the station's challenge
15	Satellite	1	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
nw-101-x1	What is meant by a standalone computer?	–	worksheet, Exit questions a)
nw-101-x2	Name one advantage of a network.	–	worksheet, Exit questions b)
nw-101-x3	Name one disadvantage of a network.	–	worksheet, Exit questions c)

ID	QUESTION	MARKS	WHERE
nw-101-x4	Give two differences between a LAN and a WAN.	—	worksheet, Exit questions d)

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/nw-101.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **What is a standalone computer?** — A computer that stands on the floor rather than a desk; A computer that connects to the internet only by Wi-Fi
- **Which of these is most likely to be standalone?** — A games console playing online; A smart speaker answering questions
- **Without a network, how can you get a file onto another computer?** — Send it by email; Save it to the cloud
- **Two friends pass one essay back and forth on a USB stick. What is the** — The file turns into a virus; The essay gets shorter every time it is copied
- **A room has 30 standalone computers. How many printers are needed so th** — 1; 3

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- What a standalone computer is
- The difference between a LAN and a WAN
- The advantages of networking
- The disadvantages of networking

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is nw-102, Network control room. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute; the experiences are used on the website.

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